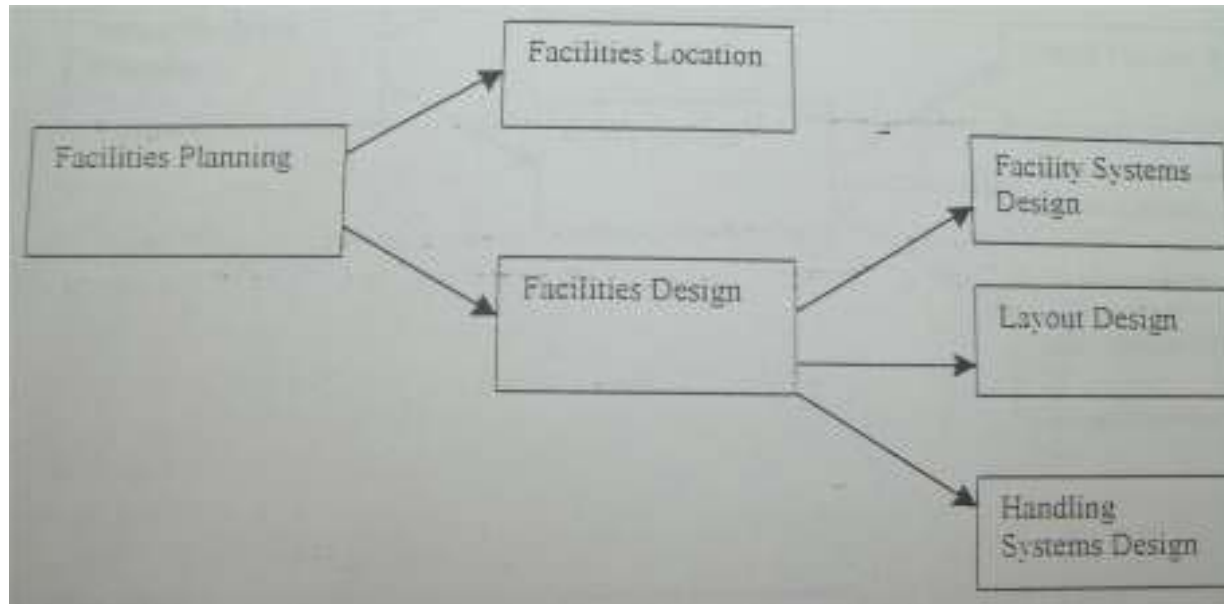


Facilities Planning $\neq$ Facility Location
Facilities Design
Facilities Layout
Plant Layout

Facilities Planning Hierarchy:



- **Facilities Location (Macro Aspect of FP):**

Location of the facility refers to its placement with respect to customers, suppliers, and other facilities with which it interfaces.

- There are many analytical techniques that can be used in facility location decisions. Some of these are:
 - *1) Factor Rating (Detail at lab)*
 - *2) Cost-Profit-Volume analysis*
 - *3) Center of Gravity Method, etc*

Facilities Design (Micro Aspect of FP):

Design components of a facility consists of

- **Facilities Systems:**

Consists of the structural systems, the atmospheric systems, the

lighting/electricity/communication systems, the life safety systems and the sanitation systems.

- **Layout:**

Consists of all equipment, machinery and furnishings within the building.

- **Handling Systems:**

Consists of the mechanism need to satisfy the required facility interactions.

- **For a manufacturing system:**
- **Facility Systems** – the structure (of building), power, light, gas, heat, ventilation, air-conditioning, water and sewage needs.
- **Layout** – the production areas, related support areas, personnel areas.
- **Handling Systems** – the materials- personnel, information, and equipment to support manufacturing.

1.1.7. Application of FP Hierarchy

- Hospital Facility Planning.
- Emergency Room Facility Planning
- Food Chain Facility Planning.

(Draw by your self, class activity)

1.2. Significance of Facilities Planning

To understand the significance of Facilities Planning (FP) consider the following questions:

- What impact does facilities planning have on handling and maintenance cost?
- What impact does facilities planning have on employee morale, and how does employee morale impact operating costs?
- In what do organizations invest the majority of their capital, and how convertible is their capital once invested?
- What impact does facilities planning have on the management of a facility?
- What impact does facilities planning have on facility's capability to adapt to change and satisfy future requirements?

1.3. Objectives of Facilities Planning

Objectives of FP is to plan a facility that achieves both facilities location and design objectives.

1.3.1. Objectives of Industrial Facility Location:

Objective of Industrial Facility Location is to determine the location which, in consideration of all factors affecting deliver-to-customers cost of the products to be manufactured, will be minimized.

1.3.2. Some Typical Facilities Design Objectives are to:

1. Support the organization's vision through improved material handling, material control, and good housekeeping.
2. Effectively utilize people, equipment, space and energy.
3. Minimize capital investment.
4. Be adaptable and promote ease of maintenance.
5. Provide for employee safety and job satisfaction.

1.4. Facilities Planning Process

- FP is not an exact science, but it can be approached using an organized and systematic approach.
- ENGINEERING DESIGN PROCESS (EDP)

In EDP we had following steps.

1. Define the problem,
2. Analyze the problem,
3. Generate alternative designs,
4. Evaluate the alternatives,
5. Select the preferred design,
6. Implement the design.

- Applying the EDP to FP results in the following process:
 1. Define (or redefine) the objective of the facility,
 2. Specify the primary and support activities to be performed in accomplishing the objective in terms of:
 - Operations, • Equipment,
 - Personnel, • Material flows should be satisfied.
 3. Determine the interrelationships among all activities,
 4. Determine the space requirements for all activities,
 5. Generate alternative facilities plans
 6. Evaluate alternative facilities plans (alternative locations and alternative designs)
 7. Select a facilities plan,
 8. Implement the facilities plan,
 9. Maintain and adapt the facilities plan,
 10. Redefine the objective of the facility.

An Organization's Model of Success:

- Five elements that form an organization's model of success:

Vision: a description of where you are headed.

Mission: how to accomplish the vision.

Requirements of success: the science of your business.

Guiding principles: the values to be used, while pursuing the vision.

Evidence of success: measurable results that will demonstrate when an organization is moving towards their vision.

TOPIC DISCUSSED:

- 1.1~1.4
- FP determines how an activity's tangible fixed assets should contribute to meeting the activity's objectives.
- FP consists of facilities location and facilities design.
- Partly art, partly science.
- Can be approached using the engineering design process.
- Represents one of the most significant opportunity for cost reduction and productivity improvement.